Evaluation based Approaches for Liver Disease Prediction using Machine Learning Algorithms

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Abstract— **The life of humans living without liver tumors is one of the fundamental care of human livelihood. Therefore, for better care, detection of liver disease at a primitive phase is necessary. For medical experts, predicting the illness in the early stages due to subtle signs is a very difficult task. Many, when it is too late, the signs become evident. The current work aims to augment the perceive nature of liver disease by means of machine learning methods to solve this epidemic. The key purpose of the present work focused on algorithms for classification of healthy people from liver datasets. Centered on their success variables, this research also aims to compare the classification algorithms and to provide prediction accuracy results.**

Keywords- Human life, classification algorithm, Linear Regression, SVM, Machine Learning

I. INTRODUCTION

The scale of patient medical records increases day by day in the health care sector. Data mining is the method of using a computer-based information system (CBIS), using modern tactics, to uncover insights from data. The machine learning method is close to that of data mining. Algorithms in machine learning can be differentiated from either supervised or unsupervised methods of learning. For statistical modelling, supervised learning approaches are commonly used. Predictive modelling is a subset of the area of clinical and business intelligence that is used to identify health risks and also to forecast individuals' potential health status. In order to store large-scale information on patient outcomes, procedures, etc., electronic health records (EHR) are used. The data on the EHR can be organised or unstructured. Electronic health records are stored in a standardised data format using managed language to log patient knowledge as a written texts that is hyperlinked in existence. The EHR aims to streamline knowledge about the clinical workflow. Ensemble learning is a well-known method used for prediction by integrating multiple ensemble models of machine learning[1]. Aggregations of various classifiers are J48, C4.5 and Naive Bayes, etc. [2]. Ensembles search for better outcomes than all of the simple classifiers. The proposed work aims to enhance the predictive and classification quality of healthcare data by developing a hybrid predictive classifier model using the classifier ensemble [3][4].

Major issues deliberated on patients with liver disease are not readily detected at starting phase since that can usually operate even though it is partly impaired. An early detection of liver disorders will improve the survival rate of the patient. There is a high probability of liver failure among Indians. India is expected to become the World Capital of Liver Diseases by 2025. Because of the deskbound lifestyle, increased alcohol intake and smoking, the pervasive prevalence of infection inside liver in India is contributing around 100 forms of liver infections are present. It would also be of great value in the medical field to build a computer that will increase the diagnosis of the disease. This systems can help doctors make correct treatment choices, and the patient queue will also be minimised by liver specialists such as endocrinologists assisted by Automated categorization Methods for Disorders in Liver part. In medical diagnosis and disease prediction, classification techniques are widely common. Michael J Sorich [5] described on chemical datasets, the classification (SVM) provides better prediction results. Paul R Harper [6] stated that an absolute greatest categorization method does not provide forecasted results. However, the unsurpassed achieving algorithm depends on distinct features of the dataset being evaluated.

II. RELATED WORKS

A number of network models based on neural have been developed in recent research assisting the physicians of liver diagnosis in medicare field, such as diagnostic support system [7], the expert system, the perceptive diagnostic model, and the hybrid recommendation framework. Furthermore, Christopher N. [8] suggested a system for the diagnosis of medical diseases, taking into account six benchmarks: liver, hepatitis heart, diabetes, breast and lymph disorders. The researchers developed WSO and C4.5-based systems, with a precision of 64.60 percent with 19 liver disorder dataset rules and 62.89 percent with 43 WSO and C4.5 rules, respectively.

In evaluation of identified categorization techniques like naive Bayes classification, C4.5, back dissemination bayesian structure, KNN with facilitate vector, Ramana[9] also conducted an acritical study on the diagnosis of liver diseases. On the Naïve Bayes classifier, the authors gained 51.59 percent accuracy, 55.94 percent on the C4.5 algorithm,

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66.66 percent with respect to BPNN, 62.6 percent with respect to Knowledge discovery and sixty two percent accurateness with respect to vector machine support algorithm.

Underperformance of liver disorder dataset in training and testing resulted from an inadequate dataset. Sug[10] has therefore developed a system related on resampling inside negligible groups in order to effectively accommodate through data which is not sufficient in nature. In regard with study paper , the author considered two decision tree algorithms like CART and C4.5[11]. Hence for required experiments, liver disorder dataset of BUPA have been perceived.

Biomedical related field is one of the most of significant fields in which mining of data is used in various researches. Although the respective area of biomedical field is concerned with lifa of human, and particularly sensitive. Numerous researches have been conducted on number of parameters for liver disease by means of mining of data in recent years. Focusing more closely at the recent studies in this area, especially in the field of biomedicine, researchers can see a range of tasks which have been used by data mining in regard of patient details, anticipation and conduct[12]. Accuracy and pace are two significant considerations in biomedical research that should be regarded primarily in coping with any illness. Data mining methods may be of considerable assistance to physicians in this respect.

With progress, a variety of computers have entered our livelihoods. Intelligence mining on the abundance help through machine learning. It is one of the most popular places where computers can be useful as the most commonly used devices. As depicted in Figure 1 the two main approaches for machine learning are supervised and unsupervised learning [13]. The goal of the below shown methods focused to be trained through the approaches of data mining and for using some training data part and another segment for evaluation.

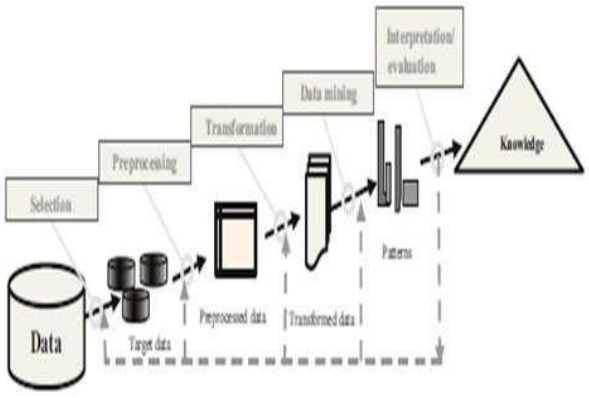


Figure.1 Steps of KDD process [14]

III. METHODOLOGY AND CLASSIFICATION ALGORTHMS

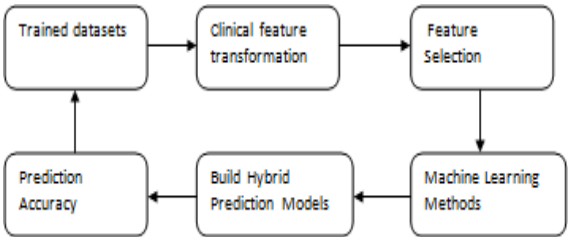


Figure 2 Pre-processing flow of proposed method

In analysis of 583 patients, Dataset of Indian Liver Patient consisted of ten distinct parameters. Based on liver disease, the patients have been characterised as either 1 or 2. Users can see a comprehensive dataset report in Table 1. This table information provides elements of attributes. All the characteristics except sex are genuine valued operands, as clearly visible from the table. In the data pre-processing step, the Sex feature is converted to an integer value (0 and 1). Figure 2 depicts the overall process of proposed system. It starts from collection of data sets, transformation process using csv formats, feature selection to apply the machine learning algorithms, building the models and finalize the prediction accuracy for liver disease prediction

Table-1 Dataset Description

No.	ATTRIBUTES	ATTRIBUTE TYPE
1.	Age	Numeric
2.	Sex	Nominal
3.	Total Bilirubin	Numeric
4.	Direct Bilirubin	Numeric
5.	Alkaline Phosphatase	Numeric
6.	Alamine Phosphatase	Numeric
7.	Total Proteins	Numeric
8.	Albumin	Numeric
9.	Albumin and Globulin Ratio	Numeric
10.	Result	Numeric (1,2)

SVM is a technique of supervised learning pertained for classification as well as regression. It has effective performance in generalisation. Moreover, when the algorithm requires input space with high dimension, there is no requirement to add a previous understanding. This helps make it a very effective classifier for quality. The primary purpose of the SVM

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classifier is to classify between groups of various classifications by choosing the best classifier function in the training data. A generalised linear method of classification is SVM. At the same time, the geometric margin is maximised and the classification error is minimised[15]. A separate hyperplane will be constructed by SVM to outlook contribution data into multi sets of vectors through x-measurement space, maximising the margin between two data sets. In order to calculate the margin, two parallel hyperplanes are constructed, one on each side of the hyperplane being separated. The greatest detachment of both the classes belonging to nearby information access facilitates for achieving separation of effective parameters. The generalisation error will decrease if the margin is large. The attribute values and scaling facilitates to obtain separating hyperplane. The following factor of data consideration is in the form of Equation 1.

$$\{(z_1, x_1), (z_2, x_2), (z_3, x_3), (z_4, x_4), \dots\dots, (z_i, x_i)\} \quad (1)$$

Zi = 1/-1 shown as stable here that provides the group in which zi fit in, and the figures due observations is i. Zi reflects the true x-measurement template. The help provided by separation with respect to hyperplane, the testing of data will be intensively perceived, as shown in Equation 2.

$$u.z + c = 0 \quad (2)$$

Where,

u is x-measurement vector and c is scalar.

The concurrent hyperplane has been selected when training data is segregated linear fashion, as there are no locations between them. The spacing seen between macro or hyper planes, that is 2/|u|, can be located geometrically here. Hence the reason figures it is necessary in reduction of |u|. Equation 3 is shown below.

$$u.z_j - c \geq 1 \text{ or } u.z_j - c < -1 \quad (3)$$

One of the simpler models for classification is logistic regression. It can be interpreted to some extent due to its parametric nature by focusing at the characteristics that make it useful when experimenters choose to focus at associations between variants. A range of vector parameters allocated from the limit (0 , 1 ... p) can fully describe model based on parametric schedule. A straight-line equation 4 as shown below.

$$Y = kx + m \quad (4)$$

Here k and m parameters would be an example of a parametric model. The complete model is envisioned using known parameters. Logistic regression is an model based on a parameters in which the characteristics are correlation coefficient written as 0 + 1 + X1 + ... PXp for the regression model. Here the value 0 is known as capture.

However this model usually focused to identify constant latent variables, the identity given ad logistic regression is less fortunate, whereas the independent variables has been discrete in classification. The term can be driven by the fact that the present work discovered the evaluation on receiptable variable holding to positive group in logistic regression, and the found expected value can be consistent.

IV. EVALUATION PARAMETERS AND RESULTS

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The main objective for this work is to predict tumor or disease in liver by means of different techniques with respect to mining techniques. The prediction is analysed by the techniques such as SVM and Logistic Regression. and all outcomes has been depicted in table 2. The parameters such as specificity ,sensitivity, precision, and accuracy are shown with each algorithm. The evaluation formula can be defined as shown in Equation 5,6 and 7[15].

$$Accuracy = \frac{no.of\,TP + no.of\,TN}{no.of\,TP+FP+FN+TN} \quad (5)$$

Precision holds the fraction of true positives defensive of measured results (combined factors of true positives (TP) and false positives (FP)) is defined as

$$Precision = \frac{no.of\ TP}{no.of\,TP+FP} \quad (6)$$

The True negative (TN) stage is shown by Specificity, which holds the measure on tuples of negative data.

$$Specificity = \frac{no.of\ TN}{no.of\,TN+FP} \quad (7)$$

Table 2 Classification Algorithm results

Classification Algorithm	**Accuracy**	**Precision**	**Sensitivity**	**Specificity**
Logistic Regression	73.23	78.57	88	30.62
SVM	75.04	77.09	79	71.11

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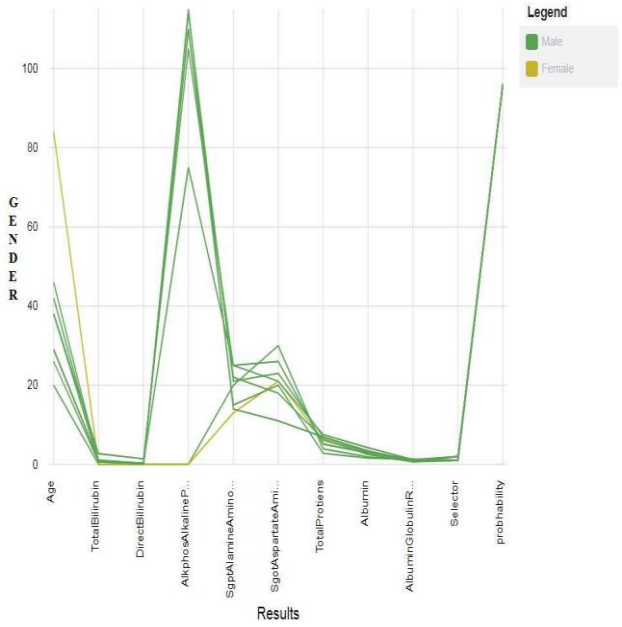


Figure 3 Probability Accuracy

V. CONCLUSION

In this article, using machine learning techniques, the methods for diagnosing liver disease in patients has been proposed and evaluated . SVM, Logistic Regression, comprises two main machine learning techniques used. Using all the models, the prediction analysis has been implemented and their performance has been assessed. The probability of liver disease prediction attained with an accuracy of 96%.In future , the present scenario can be compared with other techniques such as naïve bayes classification, Random forest etc. Also this work can be further focused on implementation of parametric classifications by bio-inspired optimization algorithms.

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